

Architecture Based Approaches Supporting Flexible Design of Self-Adaptive Software

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Abstract—It is difficult to solve an optimization problem in smart devices software due to a trade-off relation between quality conditions and performance properties. Furthermore, development cost of complicated software is on the increase by fragmentation of devices into the small size. In this paper, we propose the design processes and methods reflecting diverse of flexible elements including physical features such as low-power and high-performance, co-design of hardware and software, and restricted resources usage. Our methods can establish the optimized self-adaptive design by reflecting the quality properties and performance requirements for applications working on systems, monitoring system's states in real-time, and customizing with the most proper design properties.

Keywords- *Self-Adaptive Software, Flexible Design, Architecture Based Approaches*

I. INTRODUCTION¹

Coping with variabilities happening in smart device software has depended on an experienced based decision by skilled developer usually. If developers and maintainers are exchanged, or if new functionalities and hardware interfaces are added, we have faced many difficulties for returning to the quality of the previous changes. Also, a constant level of software quality is provided and maintained by only skilled developers. Therefore, as development of smart device software is task with high level of difficulty and cost, it is necessary of process and techniques for systematic design and constructing[1]. We suggest techniques reflecting optimized adaptive properties responsible to each application among candidates recommended from collected metadata related to software variabilities. And we defined the design process including 1) establishing a target architecture, 2) setting the design properties with aimed performance by interacting with developers literately, and 3) contracting the design properties by applying regular design patterns. Our paper are based on the premise that the usage, maintenance and flexibility in incremental evolution steps is being alleviated through model based customization of self-adaptive properties among running applications on systems.

II. SELF-ADAPTIVE SOFTWARE

We think self-adaptive software can identify, decide and perform required activities appropriately in situation for software context responding to sophisticated hardware and difficult prediction for various change properties. We expect self-adaptive software can reflect sensitively the quality goals by self-recognizing changeable requirements. In particular, smart devices software has many hardware control functions and frequent change in external environment and internal elements, that's why self-adaptive software is necessary[2][3].

III. SELF-ADAPTIVE DESIGN APPROACH

We define the metadata as software properties changed by user requirements and operation environment. We can apply them to system in static and dynamic for smart device software with big variability. It is the most important to choose the proper them among trade-off relationship properties, so metadata configuration is necessary to record, reflect and re-configure for this choice problems[4][5].

Resource		Network	
Classify	Property	Classify	Property
Processor	(1) Number	HW Channel	(1) Wireless
	(2) Speed(frequency)		(2) Cable
	(3) Response time		(3) Others
	(4) Others		
Memory	(1) Size	Topology	(1) Mesh
	(2) Cache or Not		(2) Star
	(3) Shared or Not		(3) P2P
	(4) Others		(4) Bus
Power	(1) DCFS Algorithms	Capability	(1) Bandwidth
	(2) Frequency		(2) Delay time
	(3) Others		(3) Error rate
			(4) Node storage size
			(5) Others

Figure 1. Metadata of Variable Properties

Fig. 1 shows the metadata classifying the variable properties to determine the software quality in smart device. In general, the quality elements of smart device software with the parallelism between real-time and resource usage depends a lot on the properties for network management and hardware controlling. We determine the two categories for network and resource and then variable properties of sub-parts included in each category are defined as metadata. Metadata based on self-adaptive design steps for smart device software are summarized Fig. 2.

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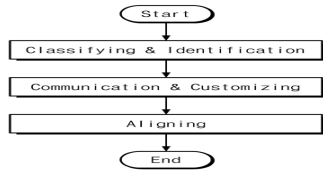


Figure 2. Summarized self-adaptive design steps

In the Classifying & Identification step, activities including inputting metadata of target smart device software, acquiring self-adaptive metadata to meet quality goals, and constructing target architecture are performed. In next step, users select and confirm repeatedly adaptive properties recommended by system. Last aligning step is for evolving into self-adaptive software. This step supports the finish of completed design by allocating and replacing the design patterns which are proved design blocks for quality goals.

Fig.3 represents the detailed self-adaptive design process of overall flow in described Fig. 2. User inputs the metadata being wanted controlling in diverse software variability and the wanted performance values for target software. And then user is recommended the best variable things from candidate properties suiting to individual software in repository being stored the diverse properties acquired by collecting the metadata. User configures architecture reflecting adaptive variable properties and considers user satisfaction of adaptive properties. If user is satisfied, proceeds next step, otherwise, user inputs new metadata into repository and recommended newly configured adaptive metadata until user are complacent. Next, completed adaptive architecture is instanced by replacing the design pattern. Structure for instating adaptive software architecture is shown in Fig. 4.

It consists of Interface, Controlling, Functions and DevicesManaging(HW controlling) part. The module for configuring metadata is in the middle. Controlling, delivery and receiving of metadata reflecting self-adaptive properties

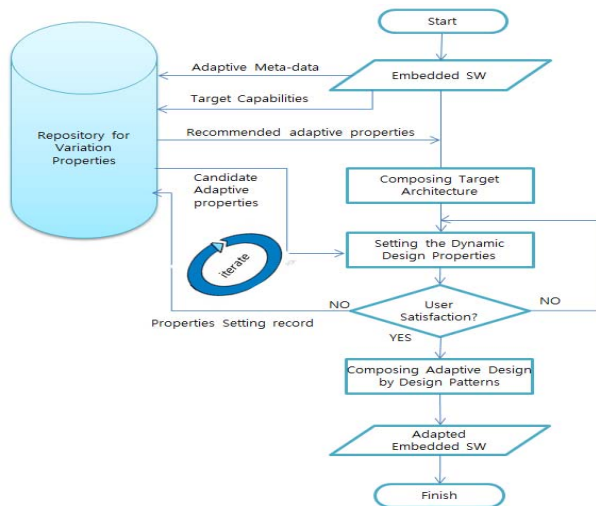


Figure 3. Self-Adaptive Design Process

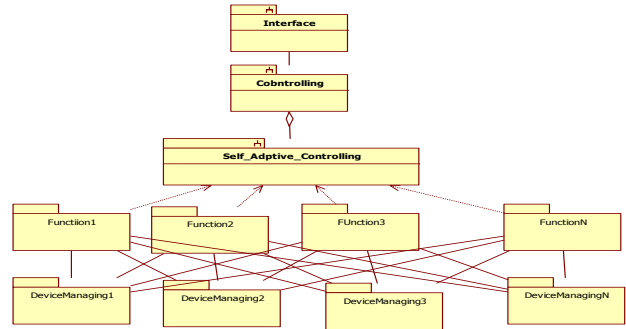


Figure 4. Self-Adaptive Design Structure(Template)

are taken charge of “Self Adaptive_Controlling” and then it contributes to controlling the properties of “Function” and “DevicesManaging” packages with dependency relation

IV. COMCLUSION

At present, self-adaptive software market for smart device is expected to considerably increase in a variety of fields. But, it's difficult to solve the optimization requirements due to trade-off relations among changeable properties. Furthermore, development cost is getting increased, because coping with the flexibility is depended on the experienced decision. For providing the self-adaptive design enhancing the flexibility of software for smart devices, we proposed the design processes and methods reflecting a diverse of flexible elements including physical features, cooperated design with hardware and software and restricted resources usage. A system development period and its cost can be saved because systems are optimized by using self-adaptive our design process. We hope our approach apply to diverse industrial filed with trade-off relation as low-price and low-power products.

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